DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

CSE2315 Lab Course
Assignment 4

- This assignment consists of several exam-level questions and sometimes an optional extra exercise.
- See Brightspace, *Course Information* for instructions about the lab course, rules and grading procedure.
- Your solution has to be handed in on paper, typeset in LaTeX, using a word processor or readable handwriting. Handing in via email is not permitted.
- Cooperating with a colleague is permitted and *strongly encouraged!* In such cases, please hand in a single copy of the work.
- Total number of pages, without this cover page: 4.

1. Suppose we have the following language over the alphabet $\Sigma = \{a, b, \#\}$:

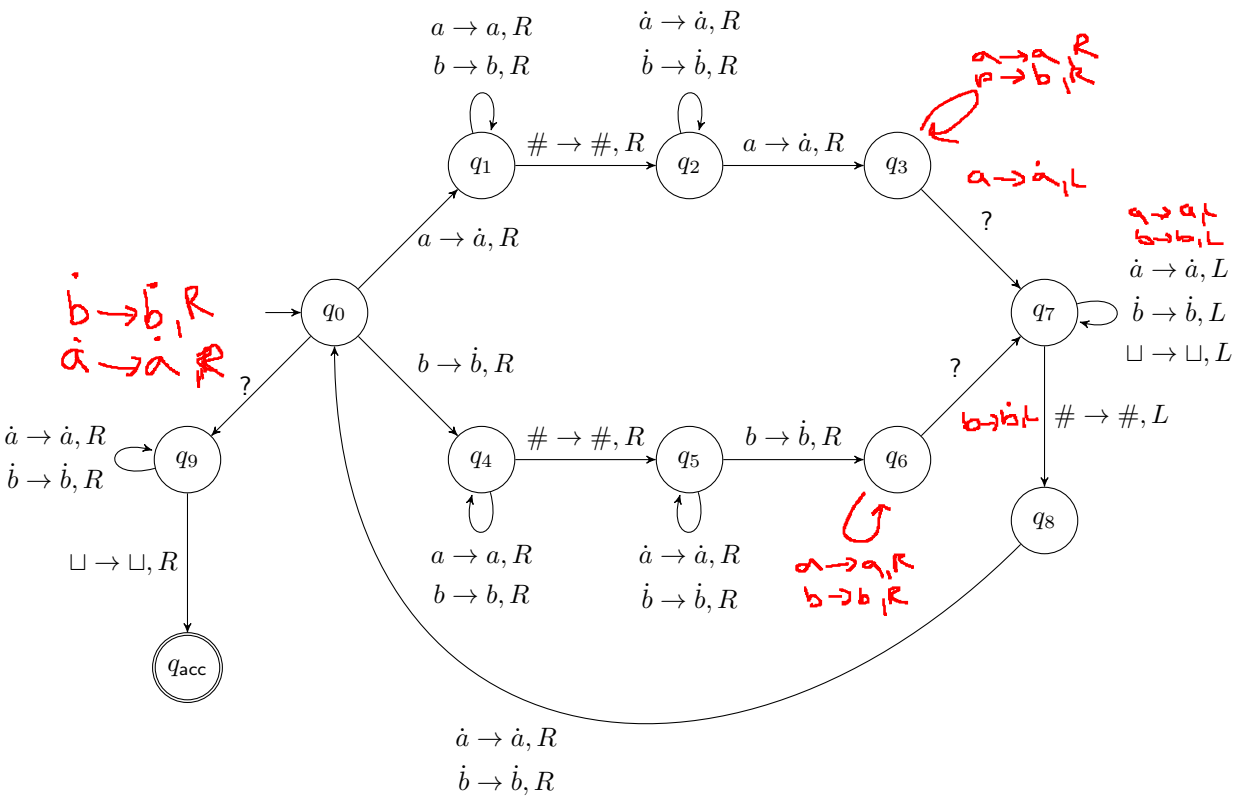
$$L = \{v\#w \mid v, w \in \{a, b\}^* \text{ and } w \text{ is } v \text{ with each letter repeated}\}.$$

Examples of words in L are:

$\#$
$ab\#aabb$
$babb\#bbaabbbb$

We have an incomplete transition diagram for a Turing machine deciding L. Assume that missing transitions lead to the reject state, with the head moving one place to the right. Question marks indicate transitions that you need to fill out in one of the questions.

NOTE: We actually missed **six** transitions for question b. We have already added one extra self-loop in q_7 the figure below.



(a) Given the above Turing machine, determine for each of the following words whether they are accepted by the machine. Consider the transitions with a question mark to be missing.

i. $a\#a$
ii. $b\#bb$
iii. $\#$

(b) Which **five** transitions need to be added such that that the Turing machine decides L? Three transitions are already marked with a question mark, the last two you have to fill in yourself. Hint: the last two transitions are loops at the same node.

2. Suppose we have the following language over the alphabet $\Sigma = \{0, 1, \#\}$:

$$L = \{0^n\#1^m \mid n > m\}$$

Give a high-level description of a nondeterministic Turing machine that decides L.

3. Suppose we have the following language over the alphabet $\Sigma = \{1, \ldots, 9, \#\}$:

$$L = \{v\#w \mid v, w \in \{1, \ldots, 9\}^* \wedge \exists a, b \in \{1, \ldots, 9\} : c_a(v) = c_b(w)\},$$

where $c_x(y)$ denotes the number of occurrences of symbol x in word y. Describe an order in which we can enumerate L. Does your order show that L is decidable?

4. Consider the following language:

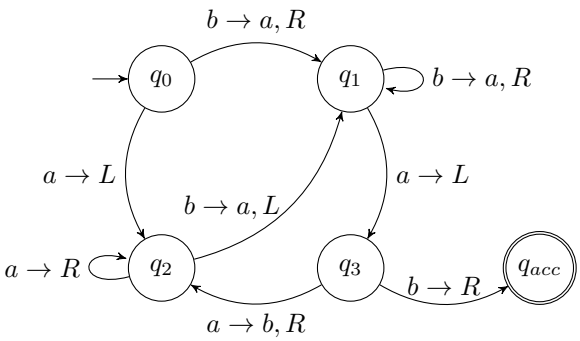
$$L = \{\langle v_1, v_2, \ldots, v_k, D\rangle \mid v_1, v_2, \ldots, v_k \in \{0,1\}^* \text{ and } D \text{ is a DFA with } L(D) = \{v_1, v_2, \ldots, v_k\}\}.$$

Is the following Turing machine M a decider for L? Explain.

$M =$ "On input $w = \langle v_1, v_2, \ldots, v_j, D\rangle$:

1. For $i = 1, \ldots, j$:
   a. Run D on v_i.
   b. If D rejects, reject.
2. If D has accepted all v_i, accept."

5. *Old exam question midterm 21-22:* Consider the following TM M_1.



(a) (1 point) What is the minimum size the tape alphabet Γ can have? Explain why.

(b) (3 points) Give a word in $L(M_1)$. Give the state transitions that show the word is in this language.

6. *Revision, old exam question endterm 22-23:* Consider the following language, where the alphabet $\Sigma = \{a, b\}$:

$L = \{w \in \Sigma^* \mid$ the maximum amount of a's following a b is equal to the maximum amount of b's following an $a\}$

For example:

- aab is not in the language as there is at most one b following an a and no as following a b.
- $aabbbaaaba$ is in the language as the largest sequence of bs following an a is size 3 and the largest sequence of as following a b is also size 3.

Consider the following start of a proof using the pumping lemma that this language is not regular.

Proof. Proof by contradiction:

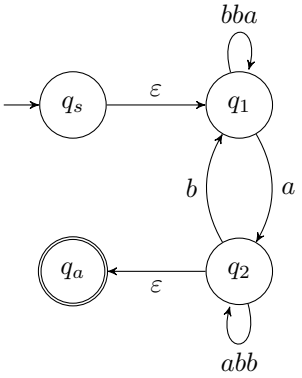
- Suppose L is regular.
- This means there must exist some pumping length $p > 0$ for L such that all words w longer than p can be split up into three parts x, y and z, with $|y| > 0$ and $|xy| \leq p$.
- For this division of w, and any $i \geq 0$, $xy^iz \in L$.
- Let's take the word $w = (a)(b)^p(a)^p$ which is in L.
- This word is longer than p, so the above holds for this word.

- ... □

Is it possible to finish the proof? If so, finish the proof. If not, explain what we would need to do to finish the proof and explain why that is not possible.

7. *Revision, old exam question midterm 20-21:*

   Given the GNFA G depicted below. If we rip q_1 out of G using the procedure described in Sipser, what (non-empty) transitions would be left in G. Make sure to describe all transitions.



See next page for an optional extra ~~exercise~~ puzzle.

Optional extra exercise This last exercise is not part of the learning material. However, if you like puzzles and have some time left, you might like to do this challenging puzzle.

Another week, another mystical mansion. This mansion seems a bit more similar to the mansion from week 1. Teleportation devices cannot teleport to multiple rooms. However, they can still teleport to 0 rooms. Nothing else is known about the mansion. This is largely because the walls block radio-signals you send or receive. Luckily, because the maximum amount of rooms that fit the mansion is still 4 and you have already made a lot of robots to use, you can just try to brute-force your way through. You can now pre-program the robot to do several instructions. If the robot isn't outside at the end of the instruction list, you let it self-destruct. Normally, if there is 4 rooms, there would always exist at least 1 order of 4 instructions that leads outside.

First, you try $Left \rightarrow Left \rightarrow Left \rightarrow Left$. When you teleport the robot inside, you can hear that the robot uses a teleportation device inside. Then, it is silent. That means that the first $Left$ instruction already lead to a teleportation device that teleports to 0 rooms. Another try, $Right \rightarrow Left \rightarrow Left \rightarrow Left$. This time, you hear the robot use 2 teleportation devices. Apparently, now the second $Left$ lead to 0 rooms. Since $Right$ seemed to do a bit better, let's immediately try $Right \rightarrow Right \rightarrow Right \rightarrow Left$. You hear the robot take 4 teleportation devices. Then, $Right \rightarrow Right \rightarrow Right \rightarrow Right$, which results in 4 teleportation devices, and then you hear the robot self-destructing. Next try, $Right \rightarrow Right \rightarrow Left \rightarrow Left$. You count the amount of teleportations devices used, and count to 6. That's strange, you gave only 4 instructions. Why didn't it already self-destruct? Let's try the last option that can be it, $Right \rightarrow Right \rightarrow Left \rightarrow Right$. Again, you count 6 teleportation devices. We are missing some information.

You suspect the teleportation devices can alter whether the robot goes to the next or the previous instruction. Also, you suspect the teleportation devices might be able to change the instruction that was just read. Given all this information, you can't be sure if there is a way to get outside. But which instructions can you try for which it is at least possible that it gets the robot outside?

Start	Left	Left	Left	Left	First teleportation device removes it
Start	Right	Left	Left	Left	Second teleportation device removes it
Start	Right	Right	Right	Left	Fourth teleportation device removes it
Start	Right	Right	Right	Right	Goes through four teleportation devices, then self-destructs
Start	Right	Right	Right	Left	Sixth teleportation device removes it
Start	Right	Right	Right	Right	Sixth teleportation device removes it

- Each room has 2 teleportation devices, a left one and a right one
- Teleportation devices are deterministic, the same device in the same room will always lead to the same room (or no room), changes the last read instruction to the same instruction, and either always lets the robot read the next instruction or always lets it read the previous instruction
- At least 1 teleportation device leads outside
- There are at most 4 rooms
- If the robot should read the previous instruction but there is none, it reads the current instruction again
- If the robot should read the next instruction but there is none, it self-destructs
- Different robots go in with instruction lists as given in the table with as an outcome the right side of the table, what is possibly a way to get the robot outside to collect the treasure?

End of the assignment